ROMAIN LANCE

Junior Mechatronics and Robotics Engineer

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Full driving license (Category B), own vehicle · **Available from December 2026**

EDUCATION

Junia HEI – École des Hautes Études d'Ingénieur (Graduate School of Engineering) – *Lille, France* 2023 – 2026
Master of Engineering (Diplôme d'Ingénieur), general engineering with a major in Mechatronics and Robotics

Junia HEI (LaSalle) – MPSI Preparatory Classes (Mathematics, Physics, Engineering Science) – *Lille, France* 2021 – 2023

PROFESSIONAL EXPERIENCE

INIT Robots – Research laboratory – *Montréal, Canada* 2026 · 6 months
Final-year internship – R&D Engineer, Mobile Robotics

- Converted a commercial agricultural robot into an open research platform running ROS 2 (ROS2)
- Designed and implemented a kinematic model driving the robot's actuators
- Designed and integrated a hardware interface between the ROS 2 architecture and the robot's CAN bus network
- Designed and integrated manual control and autonomous navigation modes

Industeam – Industrial systems integrator – *Leulinghem, France* 2025 · 4 months
Professional internship – Assistant Project Manager

- Tracked the progress of a design and manufacturing project for special-purpose industrial machines in the automotive sector
- Solved on-site issues in collaboration with the mechanical, electrical and programming teams

Splash & Fun – Water park – *In-Naxxar, Malta* 2024 · 2 months
International internship – Lifeguard

- Ensured visitor safety within an international team and customer base

Baudelet Environnement – Waste management – *Blaringhem, France* 2023 · 1 month
Blue-collar internship – Waste treatment and anaerobic digestion

- Received and cleaned organic waste containers

ACADEMIC PROJECTS

HEI Mechatronics and Robotics Research Project – Dynamic model of a smart wheelchair, CRIStAL laboratory *(team of 4 students)* 2025 – 2026

- Literature review of existing dynamic models
- Developed a dynamic model accounting for the influence of caster wheels on overall wheelchair dynamics
- Developed a model predictive control (MPC) approach
- Simulated the models in MATLAB/Simulink
- Co-authored a scientific paper

HEI Mechatronics and Robotics Project – Rover-type agricultural robot *(team of 6 students)* 2024 – 2025

- Designed and built an autonomous mobile agricultural robot
- Analysed the required actuators and sensors
- Designed the robot's mechanical parts in SolidWorks
- Programmed the robot using ROS
- Allocated tasks, led the project and managed deadlines

HEI PISTE Project – Desert refrigerator *(team of 8 students)* 2023 – 2024

- Designed and modelled a passive evaporative cooling system

TECHNICAL SKILLS

Mechanical — CAD: Fusion 360, SolidWorks · Mechanical analysis: statics, dynamics, strength of materials · Prototyping: 3D printing (FDM/PLA), basic machining, mechanical assembly

Electronics — Embedded systems: ESP32, STM32, Arduino · Communication: UART, I2C, SPI, PWM, CAN bus · Electronic design: PCB, KiCad · Platforms: Raspberry Pi, Linux

Robotics — Mobile robotics: localization, autonomous navigation, SLAM · Frameworks: ROS / ROS2 / ROS 2 · Control: open- and closed-loop control, PID and LQR control · Modelling: kinematics, dynamics, forward and inverse kinematics · Simulation: MATLAB/Simulink, Gazebo, URDF/Xacro

Programming — Languages: Python, C++, HTML · Tools: Git / version control

Languages — French: native · English: B2 (Linguaskill 161/180, TOEIC 775/990)

SOFT SKILLS AND INTERESTS

Soft skills — rigour, autonomy, adaptability, responsiveness

Interests — sports: football (club level), tennis, weight training, running · making: CAD, 3D printing, mechanics, electronics